

Ch. 27: Magnetic field

magnetic force on moving charges:

$$F = |q|vB \sin \phi \rightarrow \vec{F} = q\vec{v} \times \vec{B}$$

magnetic flux: $\Phi_B = \int B \cos \phi \, dA = \int \vec{B} \cdot \vec{A}$

* total m-flux thru any closed surface is 0: $\oint \vec{B} \cdot d\vec{A} = 0$

orbit of charged particles: when $F = \frac{mv^2}{R} \Rightarrow R = \frac{mv}{|q|B}$ (orbit radius)

from $v = R\omega \Rightarrow \omega = \frac{|q|B}{m}$ (angular velocity)

period: $T = \frac{2\pi}{\omega} = \frac{2\pi m}{qB}$

* if $\vec{v}$ is not perpendicular to $\vec{B}$ then the motion is a helix (constant $v_{||}$)

magnetic force on conductor: $J = A \cdot I$

$$\vec{F} = I \vec{L} \times \vec{B}$$

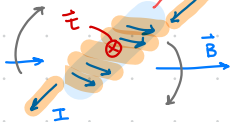
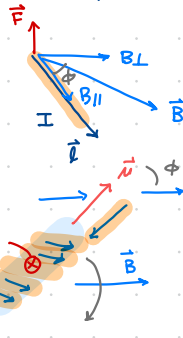
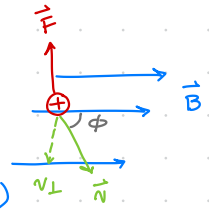
torque on current loop: total force on either side cancels out (no net force)

$$\tau = IBA \sin \phi = \mu B \sin \phi$$

magnetic dipole moment: $\vec{\mu} = I\vec{A} \rightarrow \vec{\tau} = \vec{\mu} \times \vec{B}$

* any planar loop can be approximated by rectangular loops.

potential energy of magnetic dipole: $U = -\vec{\mu} \cdot \vec{B} = -\mu B \cos \theta$

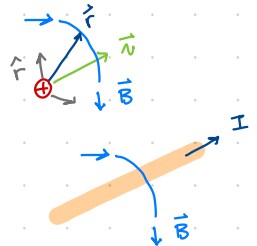


Ch. 28 Magnetic field sources

m-field from point charge: $B = \frac{\mu_0}{4\pi} \cdot \frac{|q|v \sin \phi}{r^2}$

$$\vec{B} = \frac{\mu_0}{4\pi} \cdot \frac{q\vec{v} \times \hat{r}}{r^2}$$

← unit vector from point charge to field point
↑ distance from source



m-field from current element:

Biot-Savart $\rightarrow \vec{B} = \frac{\mu_0}{4\pi} \int \frac{I d\vec{L} \times \hat{r}}{r^2}$

From Ampere's Law:

m-field of straight wire: $B = \frac{\mu_0 I}{2\pi r}$ (distance from conductor)

surface current density: $K = \sigma v$ (velocity)
↑
charge/area

infinite moving sheet of charge (belt):

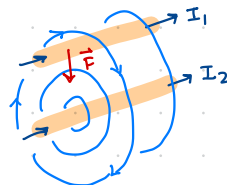
$$B = \frac{\mu_0 K}{2}$$

m-field from parallel conductors:

$$\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi r}$$

same direction $\Rightarrow$ attraction

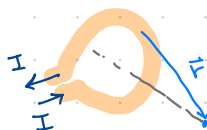
opposite direction $\Rightarrow$ repulsion



m-field from circular current loop:

on axis: $B_x = \frac{\mu_0 I a^2}{2(x^2 + a^2)^{3/2}}$

loop radius
distance along axis from center to field point



at center ($x=0$): $B_x = \frac{\mu_0 N I}{2a}$

loops

solenoid: $B = \mu_0 n I$

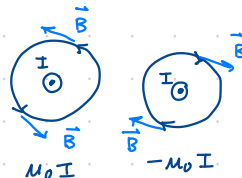
turns per unit length
(center?)

magnetic moment of N loops:

$$\mu_N = N I \pi a^2$$

Ampere's Law: $\oint \vec{B} \cdot d\vec{\ell} = \mu_0 I_{enc}$

→ ± current in RHR direction:



Ch. 29: Electromagnetic Induction

emf (voltage across a circuit element): $\mathcal{E} = \int_a^b \vec{E} \cdot d\vec{\ell}$

equivalent electric field

Faraday induction: $\mathcal{E} = - \frac{d\Phi_B}{dt} = \oint \vec{E} \cdot d\vec{\ell}$

induced emf

loops → $\mathcal{E} = -N \frac{d\Phi_B}{dt}$

RHR for direction: $\vec{A}$ (thumb), current (curl fingers)

Lenz's Law: induced emf opposes its source

changing B → changing Φ → emf $\mathcal{E}$ → current → B

OR changing A

opposes

motional emf: $\mathcal{E} = vBL$ → direction from Lenz.

speed of moving conductor

length of conductor

* m-field must be constant across entire length

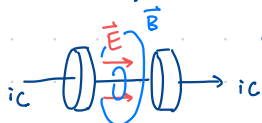
general: $\mathcal{E} = \oint (\vec{v} \times \vec{B}) \cdot d\vec{\ell}$ line integral over all elements of closed current loop

displacement current: $i_D = \epsilon \frac{d\Phi_E}{dt}$

↑ permittivity of material in area

current density: $j_D = \epsilon \frac{d\vec{E}}{dt} = \frac{i_D}{A}$

general Ampere's Law: $\oint \vec{B} \cdot d\vec{\ell} = \mu_0 (i_c + i_D)_{enc}$



a capacitor being charged by current i_c can be thought of as having a displacement current equal to i_c between the plates (as the source of the m-field between plates)

(Maxwell): $\oint \vec{B} \cdot d\vec{\ell} = \mu_0 (i_c + \epsilon_0 \frac{d\Phi_E}{dt})_{enc}$

Ch. 30 Inductance

mutual inductance: $M = \frac{N_2 \Phi_{B2}}{i_1} = \frac{N_1 \Phi_{B1}}{i_2}$ [H]
 changing I in coil 1 $\rightarrow$ changing $\vec{B}$ and Φ_B in coil 2 $\rightarrow$ emf (& current) in coil 2

mutually induced emfs: $\mathcal{E}_2 = -M \frac{di_1}{dt}$ $\mathcal{E}_1 = -M \frac{di_2}{dt}$

self-inductance of a coil: $L = \frac{N \Phi_B}{i}$
 current i in coil changes $\rightarrow$ changing Φ_B thru coil $\rightarrow$ emf induced in coil

$N \frac{d\Phi_B}{dt} = L \frac{di}{dt}$ self induced emf $\rightarrow \mathcal{E} = -L \frac{di}{dt}$ $\leftarrow$ lenz's law

inductors in circuits: $V_{\text{ab}} = L \frac{di}{dt} \Leftrightarrow V = RI$ for resistors
 voltage difference across an inductor (temporarily stores energy in magnetic field)

inductors in series:

$$L_{\text{eq}} = L_1 + L_2 + 2M$$

parallel, same curr:

$$L_{\text{eq}} = \frac{L_1 L_2 - M^2}{L_1 + L_2 - 2M}$$

opposing curr:

$$L_{\text{eq}} = \frac{L_1 L_2 - M^2}{L_1 + L_2 + 2M}$$

magnetic field energy: $P = VI \Rightarrow P = Vi$ (instantaneous)

$$P = \frac{dE(t)}{dt}$$

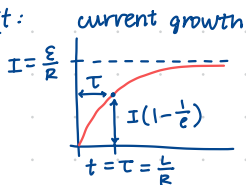
energy stored in inductor:

$$U = L \int_0^I i di = \frac{1}{2} LI^2$$

inductance $\swarrow$ $\nwarrow$ final current
 initial to final current value

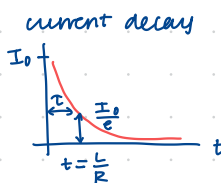
magnetic energy density: $u = \frac{B^2}{2\mu_0}$

R-L circuit:



time constant: $\tau = \frac{L}{R}$

$\hookrightarrow$ how quickly current builds towards final value



L-C circuit: oscillating current and charge, constant total energy
 thru inductor in capacitor
 m-field $\leftrightarrow$ e-field

angular frequency: $\omega = \sqrt{\frac{1}{LC}}$

underdamped: $\omega' = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$

$$i = \frac{dq}{dt} = -\omega Q \sin(\omega t + \phi)$$

total energy: $\frac{1}{2} Li^2 + \frac{q^2}{2C} = \frac{Q^2}{2C}$

$$\Rightarrow i = \pm \sqrt{\frac{1}{LC}} \sqrt{Q^2 - q^2}$$

current charge in capacitor